Embry-Riddle Aeronautical University

Modeling Magnetic Alignment Dynamics in a Drone Payload Docking System

MA442: Math Methods Engr & Physics II - SPR 2026
Applying Differential Equations (ODEs and PDEs)

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Abstract

This project models the alignment of a metal docking system used on a heavy-lift payload drone. A rotational differential equation is developed from magnetic torque and damping effects. The system is analyzed and numerically solved to determine the behavior of the above systems. Results showcase how parameters affect the drone's stability and docking speed.

Introduction

This project models the dynamics of magnetic alignment for a heavy-lift payload drone. The system is designed such that components on the drone and payload, or connectors, will converge to create a secure connection, allowing the drone to continue its mission. To engage the locking mechanism during flight, magnets align the connectors once the drone is near the payload. This is ideally a smooth, quick, and reliable process.

To allow for prediction of alignment speed, stability, overall performance, and success, it is necessary to model this system using rotational dynamics and magnetic torque principles over time. This approach will provide insight into system behavior prior to testing and will support design decisions for successful takeoff.

Problem Statement

The objective of this project is to model rotational dynamics and of a magnetically guided system using magnetic torque principles over. The system assumes free rotation until the magnetic forces align the drone with the target orientation of the payload. The project model establishes how system parameters such as magnetic strength and damping influence speed, stability, and overall performance. Additionally, the model evaluates convergence and oscillatory interference with successful connection.

Governing Equations

Newton's Second Law for Rotational Motion

Newton's Second Law for rotational systems is used during the rotational motion for the alignment of the connector system. It is assumed to rotate about two independent angular coordinates in a ball-joint style. This design makes it possible for the connector component on the drone to align with the payload more efficiently before latching.

$\theta(t)$ and $\phi(t)$, where Θ = pitch and Φ = yaw, symbolize the angular displacements of the two rotational axes. We can assume motion in each axis is driven by magnetic restoring torque and resisted by damping effects. Newton's Second Law for rotational motion can be seen below.

$$I\frac{d^2\alpha}{dt^2} = \tau \text{ [1]}$$

Where: I = moment of inertia about axis of rotation, α = angular displacement, and τ = net torque acting on connector.

Magnetic Torque

Magnetic interaction on this system is to align the connector with the payload's orientation. Magnetic Torque is modeled below as a restoring torque proportional to the sine of the misalignment angle [2].

$$\tau_{mag,\theta} = -k_\theta \sin(\theta) \text{ [3]} \qquad \tau_{mag,\phi} = -k_\phi \sin(\phi) \text{ [3]}$$

Where: k_θ and k_ϕ = effective magnetic alignment strength about respective axes, the negative sign indicates torque acts to reduce misalignment and pushes the system toward equilibrium at $\theta, \phi = 0$.

Damping Torque

During motion, resistive effects, like joint friction, air resistance, and energy dissipation are accounted for using linear damping for each axis. Damping torque is assumed to be proportional to angular velocity as shown below.

$$\tau_{damp,\theta} = -b_\theta \frac{d\theta}{dt} \text{ [3]} \qquad \tau_{damp,\phi} = -b_\phi \frac{d\phi}{dt} \text{ [3]}$$

Where: b_θ and b_ϕ = damping coefficients for each respective axis.

Combing the equations for each axis gives the second-order system below. They describe how the connector rotates over time to create mechanical alignment in two directions.

$$I_\theta \frac{d^2\theta}{dt^2} = -k_\theta \sin(\theta) - b_\theta \frac{d\theta}{dt} \qquad I_\phi \frac{d^2\phi}{dt^2} = -k_\phi \sin(\phi) - b_\phi \frac{d\phi}{dt}$$

First-Order System Form

To implement these equations, first-order system equations are displayed below as they are more convenient for numerical solutions.

$$\omega_\theta = \frac{d\theta}{dt} \qquad \omega_\phi = \frac{d\phi}{dt}$$

Governing equations now become….

$$\dot{\theta} = \omega_\theta \qquad \dot{\omega_\theta} = \frac{-k_\theta \sin(\theta) - b_\theta \omega_\theta}{I_\theta} \qquad \dot{\phi} = \omega_\phi \qquad \dot{\omega_\phi} = \frac{-k_\phi \sin(\phi) - b_\phi \omega_\phi}{I_\phi}$$

Initial Conditions

A realistic docking scenario will assume the connector will have initial angular misalignment and zero initial angular velocity.

$$\theta(0) = \theta_0 \qquad \omega_\theta(0) = 0 \qquad \phi(0) = \phi_0 \qquad \omega_\phi(0) = 0$$

Where: θ_0 and ϕ_0 = initial angular offsets from the aligned position.

Assumptions

The following assumptions will be made.

- The connector will behave as a ridged body.
- Rotation will only occur on the two-axis described.
- Magnetic torque an axis can be approximated as a sinusoidal restoring torque [2]
- Damping is linear to angular velocity
- Motion about the two axes is treated independently
- External disturbances and translational motion are neglected

Numerical Method

This project will use Euler's Method due to its simplicity. This method approximates the solution at a step time interval by using the most recent variable and its time derivative. Euler's Method is found below for a general problem.

$$y_{n+1} = y_n + hf(t_n, y_n) \text{ [4]}$$

Where: y = state variable and h = time step size

After applying this method to system, we gather:

$$\theta_{n+1} = \theta_n + h\omega_{\theta,n} \quad \omega_{\theta,n+1} = \omega_{\theta,n} + h\left(\frac{-k_\theta \sin(\theta_n) - b_\theta \omega_{\theta,n}}{I_\theta}\right)$$

$$\phi_{n+1} = \phi_n + h\omega_{\phi,n} \quad \omega_{\phi,n+1} = \omega_{\phi,n} + h\left(\frac{-k_\phi \sin(\phi_n) - b_\phi \omega_{\phi,n}}{I_\phi}\right)$$

The simulation begins at the initial conditions and with the respective angular velocities until the simulation time. A small time step is chosen so the solution is stable and accurate. $\theta(t), \phi(t), \omega_\theta(t), \text{ and } \omega_\phi(t)$ are plotted to show the magnetic strength and damping for alignment performance. For the purpose of this simulation, arbitrary values are given to moments of inertia, magnetic constants, damping coefficients, and initial angular offsets to produce realistic behavior.

Results

The simulation was done using the above method as described (Euler's Method) for the two-axis system. Initial conditions were arbitrarily selected to perform as they would in real application. All code used to calculate plots can be found in the Appendix. Results display in Figure 1 that both axes angular displacements decrease toward zero over time, showing successful connection.

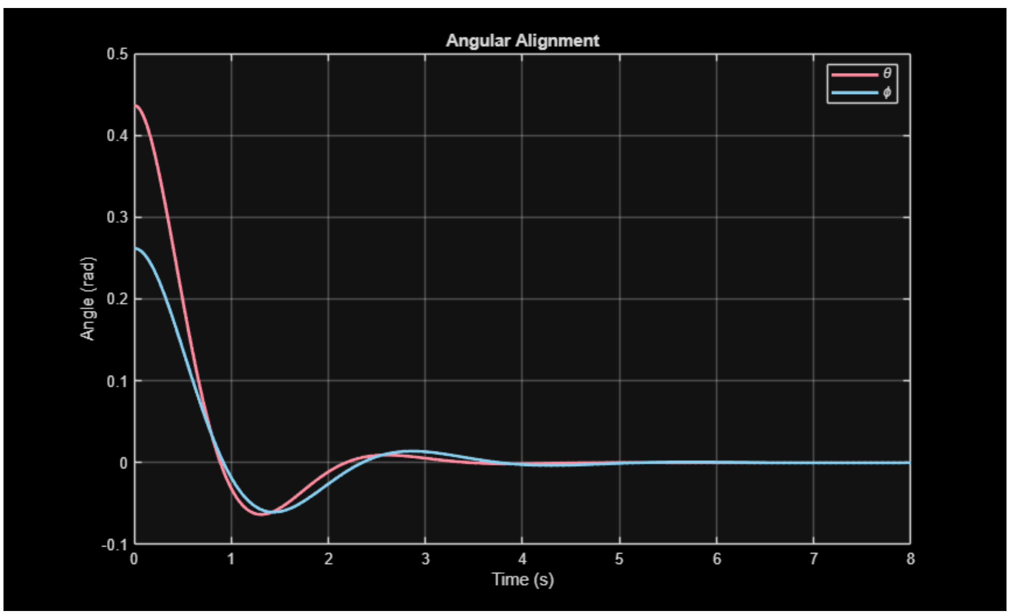


Figure 1: Angular alignment about the theta and phi axes.

The angular velocity plot in Figure 2 below displays that the connector starts by accelerating toward the equilibrium position and gradually slows as damping dissipates the gathered energy. This behavior is consistent with damped rotational alignment processes.

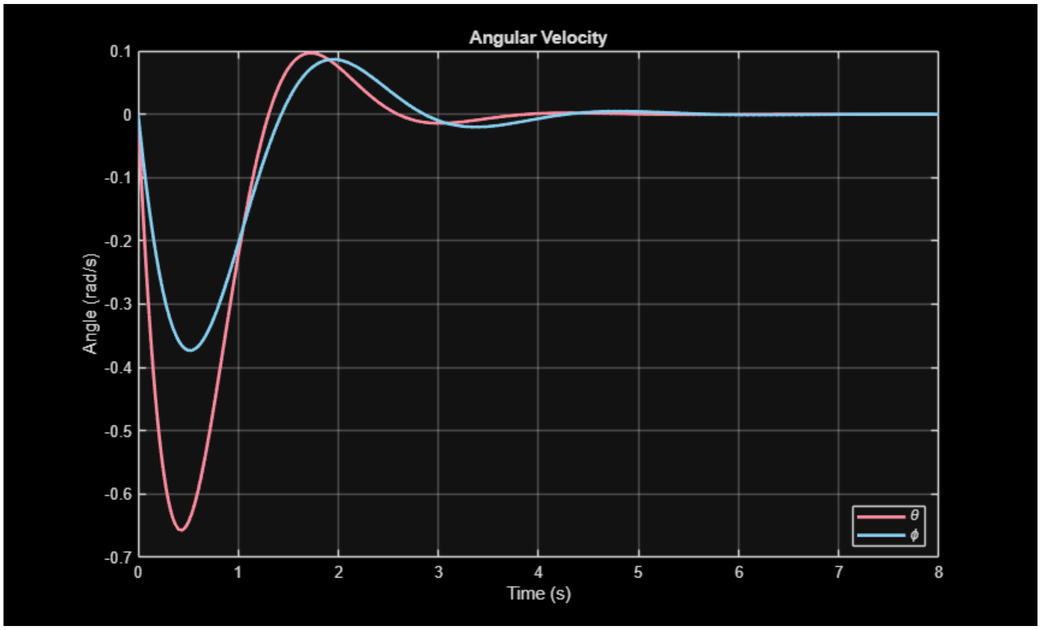


Figure 2: Angular velocity about the theta and phi axes.

Simulations to show magnetic strength and damping were also calculated. As shown is Figure 3 and 4 below, increasing the magnetic alignment constants subsequently affected the connector to align more quickly. Additionally, increasing the damping coefficients decreased oscillatory motion and produced a smoother connection.

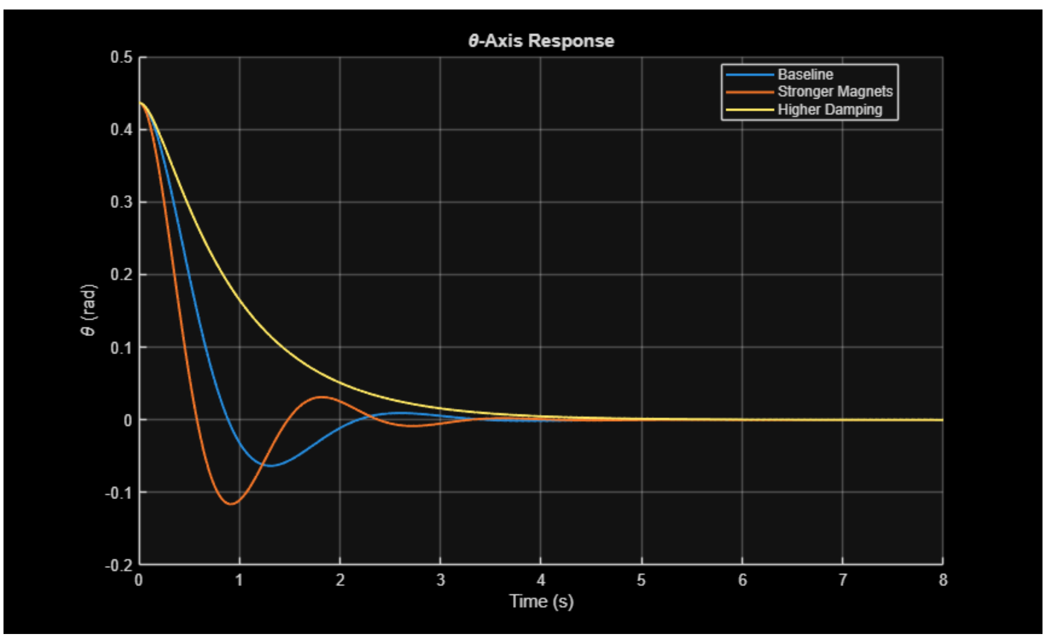


Figure 3: Theta axis response for different parameters.

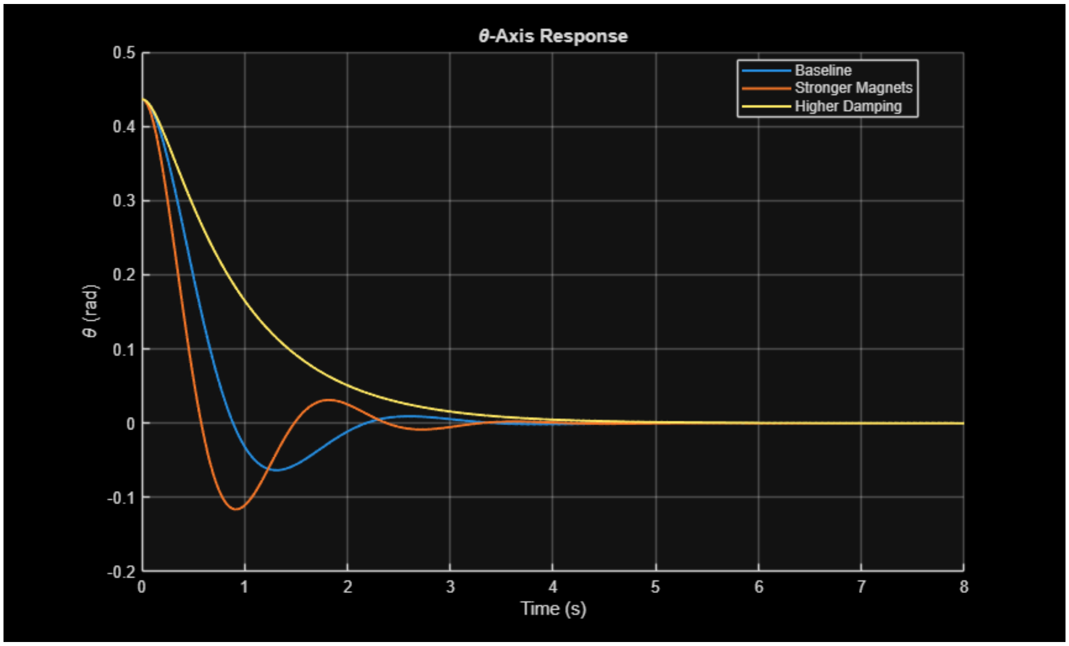


Figure 4: Phi axis response for different parameters.

Discussion

The results showcase that the magnetic alignment system can be represented as a two-axis damped rotational system. For both axes, the restoring magnetic torque moves the connector towards the aligned position with the damping variable slowing oscillation. For a docking system, large oscillations can interfere with the latching speed so this behavior is desired.

Increasing magnetic strength improved alignment speed, this suggests that stronger magnets will decrease docking time, increasing overall drone mission efficiency. However, it must be considered for the final design of the docking system that stronger magnetic torque could also produce higher angular velocities.

Increasing damping reduced oscillations, improved stability, and created a smoother alignment process. This showcases that damping is a necessary design parameter for docking.

This model or project is sufficient for heavy-lift payload drone prototyping and testing as it increases the known information about the system. Although the model assumes independent motion about each axis, the alignment for the ball-joint and connector system still showcases the behavior of system. Future research may include coupling between axes and translational motion.

Conclusion

This project modeled the magnetic dynamics of alignment of two different axes for a heavy-lift payload drone connector system using rotational differential equations. Euler's Method was applied to a nonlinear set of ordinary differential equations that applied magnetic torque and damping effects.

Results showcased that magnetic forces can drive the connector toward alignment for both axes and that damping played a critical role in decreasing oscillatory behavior for the connecting. Increased magnetic strength increased alignment speed, while increased damping effects increased stability and control.

Applying these results to the design of the connector, it can be assumed that fast and efficient alignment is necessary for a successful mission. This model will provide insight into the design phase of the connector for how system parameters influence performance. This will majorly increase prototype efficiency.

Future work may include coupled multi-axis dynamics, translational motion, and more information about a magnetic force to further improve accuracy and adaptability to real-world flight and mission conditions.

References

[1] R. C. Hibbeler, Engineering Mechanics: Dynamics, Pearson (October 21, 2021) © 2022.

[2] D. J. Griffiths, Introduction to Electrodynamics, Upper Saddle River, New Jersey 07458: Prentice Hall, 1999.

[3] S. H. Strogatz, Nonlinear Dynamics and Chaos with Appllications to Physics, Biology, Chemistry, and Engineering, 6000 Broken Sound Parkway NW, Suite 300: CRC Press Taylor & Francis Group, 2018.

[4] D. G. Zill, Advanced Engineering Mathematics, 6th Edition ed., 5 Wall Street Burlington, MA 01803: Jones & Bartlett Learning, 2017, p. 71.

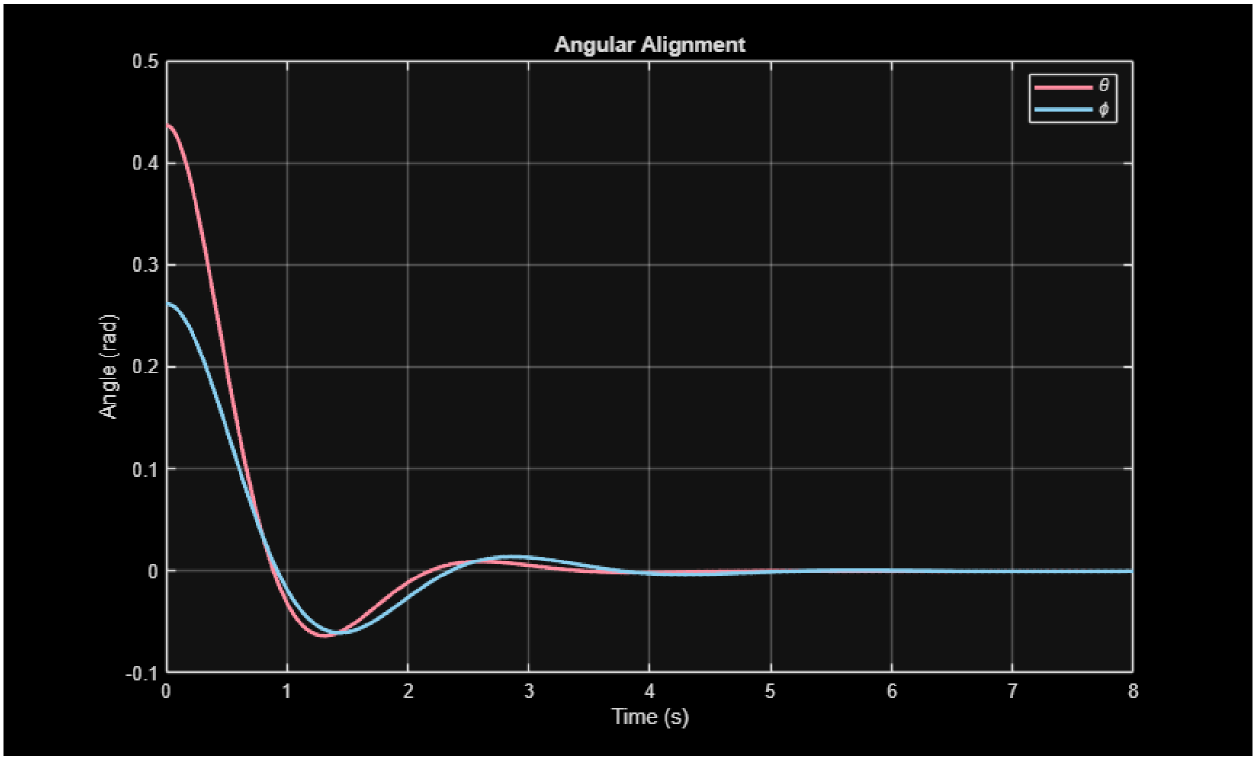
Appendix

Note: All code was ran in MATLAB.

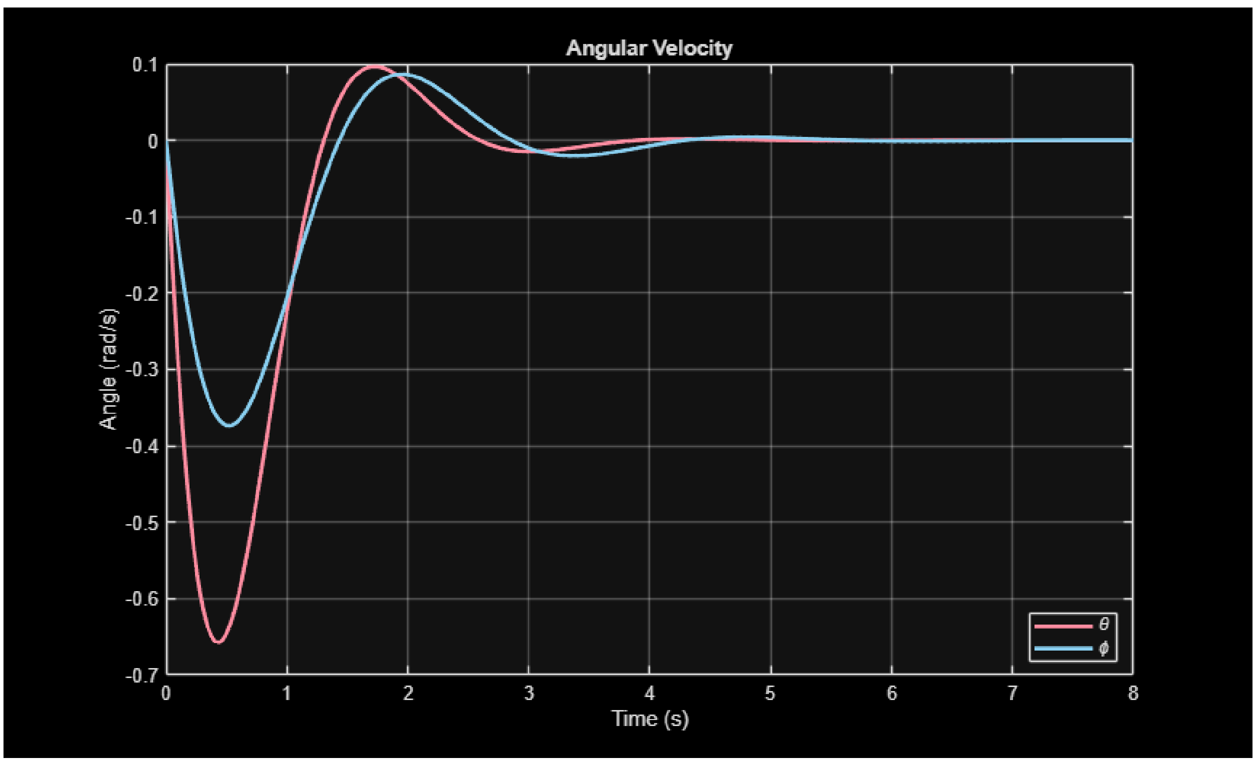
```matlab
% MA442
% Applying Differential Equations (ODEs and PDEs)
% Modeling Magnetic Alignment Dynamics in a Drone Payload Docking System
% Clancy Crawford   Student ID:2654296
% 4/10/26

clear;
close all;
clc;

% Parameters
I_theta = 0.01; %(kg*m^2)
I_phi   = 0.012;
k_theta = 0.08; % magnetic alignment strength
k_phi   = 0.07;
b_theta = 0.03; % damping coefficient
b_phi   = 0.025;
theta0 = deg2rad(25); % initial misalignment [rad]
phi0   = deg2rad(15);
omega_theta0 = 0; % initial angular velocity [rad/s]
omega_phi0   = 0;

% Time function
h = 0.01; % time step in seconds
t_final = 8; % final simulation time
t = 0:h:t_final;
N = length(t);

% make arrays
theta = zeros(1,N);
phi = zeros(1,N);
omega_theta = zeros(1,N);
omega_phi = zeros(1,N);

% ICs
theta(1) = theta0;
phi(1) = phi0;
omega_theta(1) = omega_theta0;
omega_phi(1) = omega_phi0;

% Euler's Method
for i = 1:N-1
    % Angular accelerations
    alpha_theta = (-k_theta*sin(theta(i)) - b_theta*omega_theta(i))/I_theta;
    alpha_phi   = (-k_phi*sin(phi(i))   - b_phi*omega_phi(i))/I_phi;

    % Update angular velocities
    omega_theta(i+1) = omega_theta(i) + h*alpha_theta;
    omega_phi(i+1)   = omega_phi(i)   + h*alpha_phi;

    % Update angles
    theta(i+1) = theta(i) + h*omega_theta(i);
    phi(i+1)   = phi(i)   + h*omega_phi(i);
```

```
end

% angluar alignment
figure;
plot(t, theta, 'Color', '#FF8DA1', 'LineWidth',2);
hold on;
plot(t, phi, 'Color', '#89CFF0', 'LineWidth', 2);
grid on;
xlabel('Time (s)');
ylabel('Angle (rad)');
title('Angular Alignment');
legend('\theta', '\phi', 'Location','best')

% angular velocity
figure;
plot(t, omega_theta, 'Color', '#FF8DA1', 'LineWidth', 2);
hold on;
plot(t, omega_phi, 'Color', '#89CFF0', 'LineWidth', 2);
grid on;
xlabel('Time (s)');
ylabel('Angle (rad/s)');
title('Angular Velocity');
legend('\theta', '\phi', 'Location','best')

% Comparing Cases
cases = {
    struct('name','Baseline','k_theta',0.08,'k_phi',0.07,'b_theta',0.03,'b_phi',0.025), ...
    struct('name','Stronger Magnets','k_theta',0.14,'k_phi',0.12,'b_theta',0.03,'b_phi',0.025),
...
    struct('name','Higher Damping','k_theta',0.08,'k_phi',0.07,'b_theta',0.08,'b_phi',0.07)
};

theta_all = zeros(length(cases), N);
phi_all = zeros(length(cases), N);

% initalizing comparison results
for c = 1:length(cases)
    theta = zeros(1,N);
    phi = zeros(1,N);
    omega_theta = zeros(1,N);
    omega_phi = zeros(1,N);

    theta(1) = theta0;
    phi(1) = phi0;
    omega_theta(1) = omega_theta0;
    omega_phi(1) = omega_phi0;

    for i = 1:N-1
        alpha_theta = (-cases{c}.k_theta*sin(theta(i)) -
cases{c}.b_theta*omega_theta(i))/I_theta;
        alpha_phi   = (-cases{c}.k_phi*sin(phi(i))   - cases{c}.b_phi*omega_phi(i))/I_phi;

        omega_theta(i+1) = omega_theta(i) + h*alpha_theta;
        omega_phi(i+1)   = omega_phi(i)   + h*alpha_phi;
```

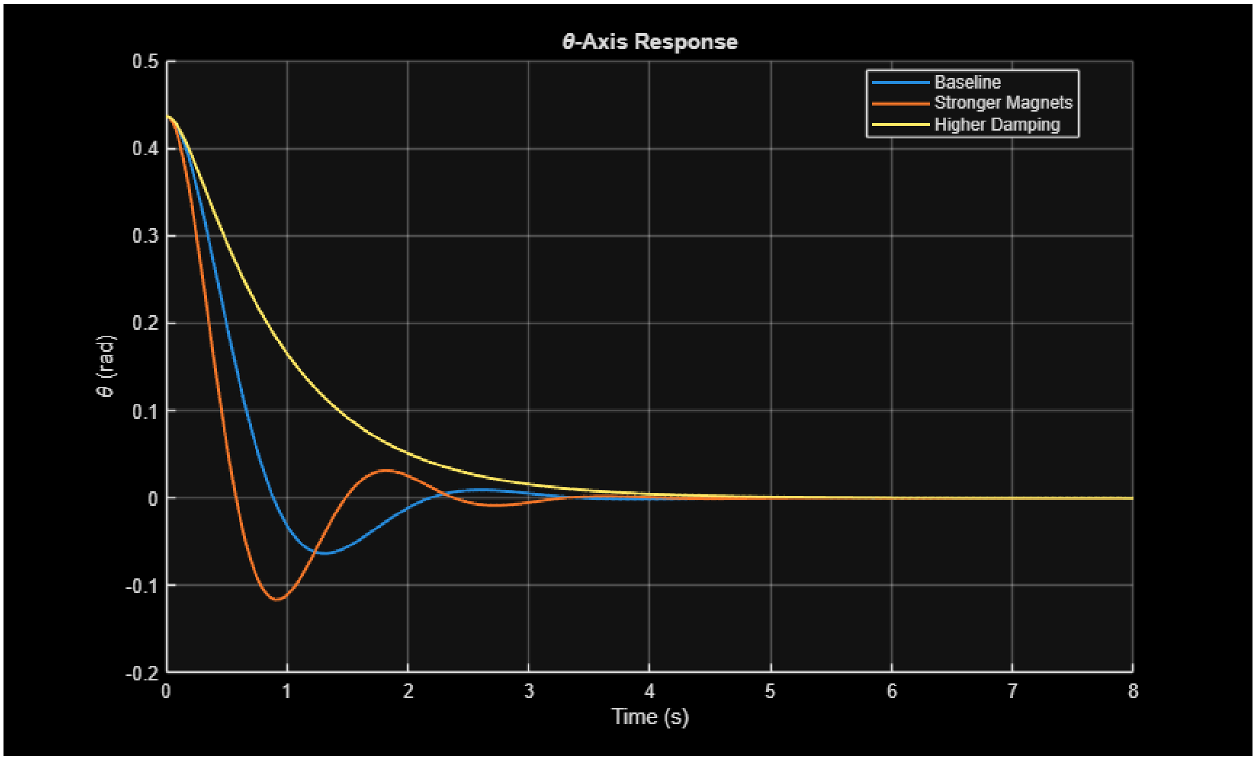
```matlab
        theta(i+1) = theta(i) + h*omega_theta(i);
        phi(i+1)   = phi(i)   + h*omega_phi(i);
    end

    theta_all(c,:) = theta;
    phi_all(c,:) = phi;
end

% Comparison plot for theta
figure;
hold on;
for c = 1:length(cases)
    plot(t, theta_all(c,:), 'LineWidth', 1.5, 'DisplayName', cases{c}.name);
end
grid on;
xlabel('Time (s)');
ylabel('\theta (rad)');
title('\theta-Axis Response');
legend('Location','best');

% Comparison plot for phi
figure;
hold on;
for c = 1:length(cases)
    plot(t, phi_all(c,:), 'LineWidth', 1.5, 'DisplayName', cases{c}.name);
end
grid on;
xlabel('Time (s)');
ylabel('\phi (rad)');
title('\phi-Axis Response');
legend('Location','best');
```

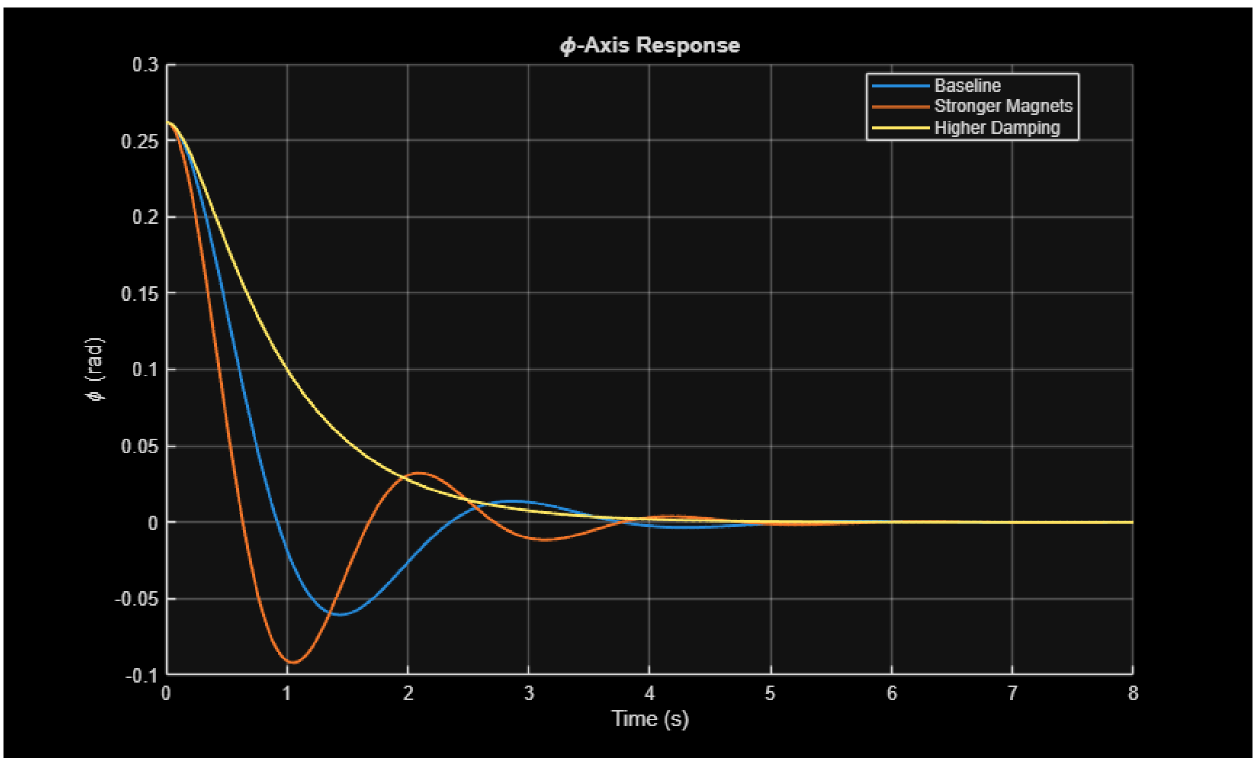


Angular Alignment
θ
ϕ
Angle (rad)
Time (s)



Angular Velocity
θ
ϕ
Angle (rad/s)
Time (s)





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